

BS2 Series B • Supplement

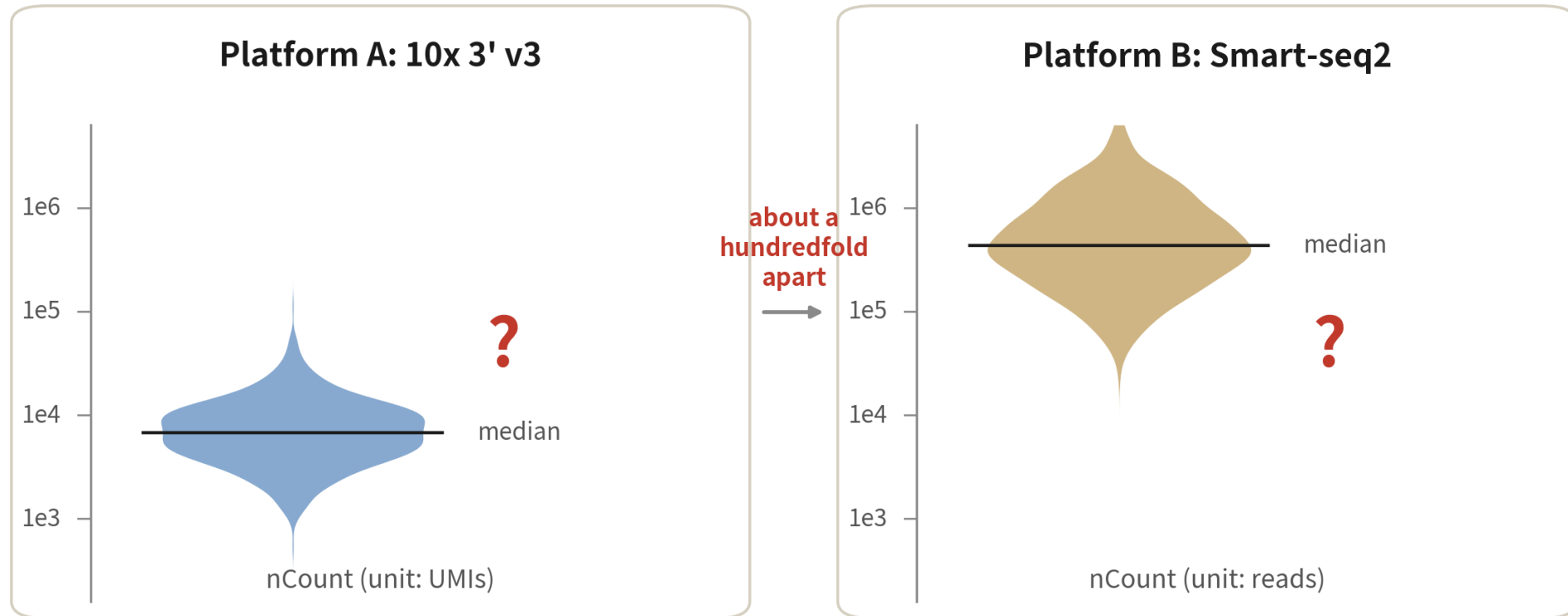
The platform landscape: life beyond 10x

Droplets, microwells, plates, combinatorial barcoding — all four families in one sitting, and how the platform dictates your QC and analysis.

About 35 min | Prerequisites: B2-B5 | Series B Supplement, episode 2

Same blood, two platforms — wildly different QC

Same tube of blood, same day, two machines — the QC reports are back



Which machine is broken? — hold that thought

simulated data (positions indicate orders of magnitude)

The two QC plots sit about a hundredfold apart — which machine is broken?

WHERE THIS EPISODE STARTS

**Neither machine is broken.
Change the platform, and 'normal' changes
shape.**

One counts UMIs, the other counts reads — not even the same unit. If you can't read the platform, you can't read the QC.

What you will learn

- 1 Place any single-cell platform in its family with three questions: how cells are isolated, whether there are UMIs, and which part of the transcript is read
- 2 State each platform's cell-count scale, depth profile, and biases — and name its representative dataset in the exercise bank
- 3 Explain how the platform shifts QC expectations, normalization choices and downstream analysis — and choose a platform from the research question

01

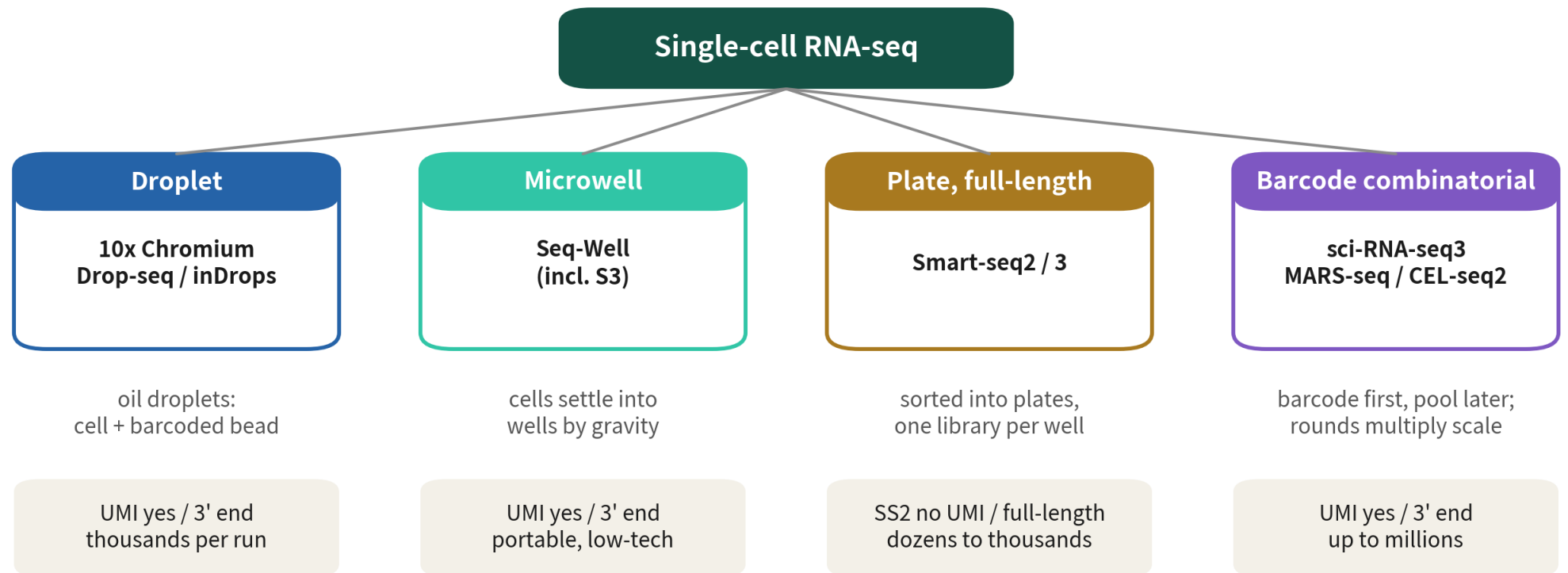
How the family tree splits

Every platform answers one question: how do you lock a cell to its barcode?

The technology family tree

The four families of single-cell technologies

One question splits the tree: how do you lock one cell to one barcode?



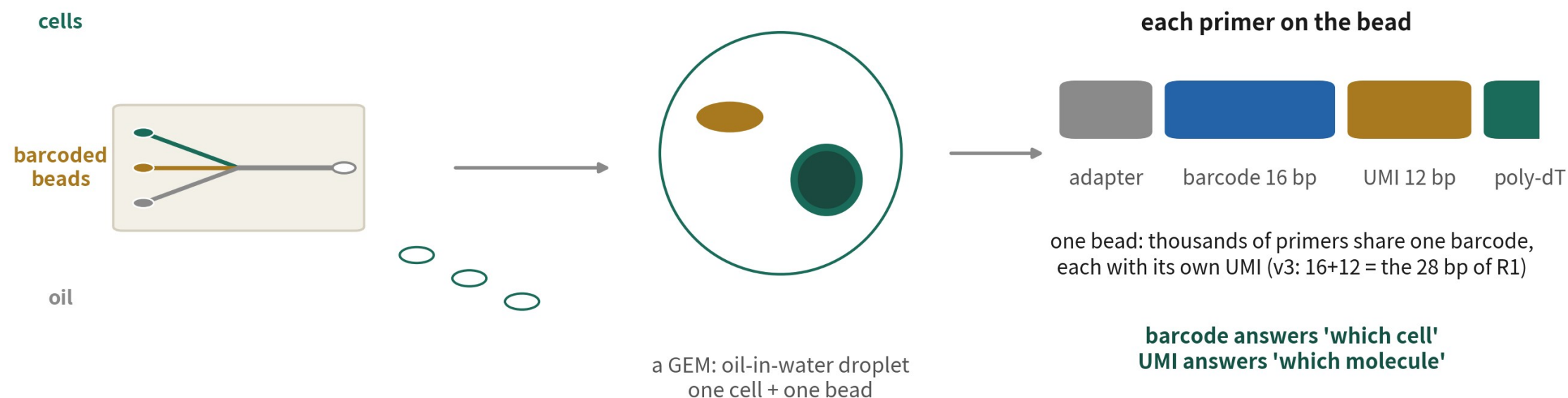
schematic (scales are orders of magnitude, from the exercise datasets)

Droplet, microwell, plate full-length, combinatorial — the difference is how isolation happens

Droplets: one drop, one isolation room

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Cell, bead and oil streams meet in a channel — every droplet is its own reactor



schematic

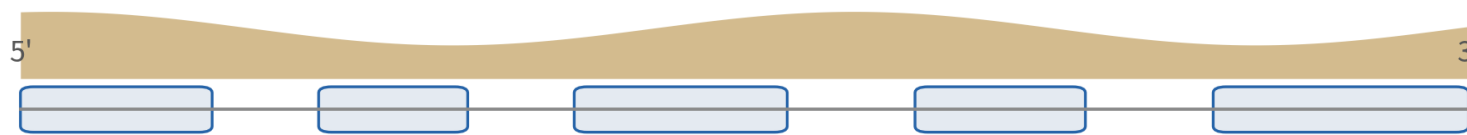
The 28 bases of R1 you dissected in B2 (16 barcode + 12 UMI) physically live on the bead's primers

What gets read: full-length vs the 3' end

What gets read: full-length vs the 3' end

Same transcript — the two platforms see different parts

Plate, full-length (Smart-seq2)



coverage tiles the whole body:
isoforms, coding mutations,
full-length TCR/BCR

Droplet 3'-end counting (10x 3')



only a stub at the 3' end:
molecules counted cleanly by UMI,
most sequence content unseen

full-length = rich but costly per cell; end-counting = clean, cheap, scalable

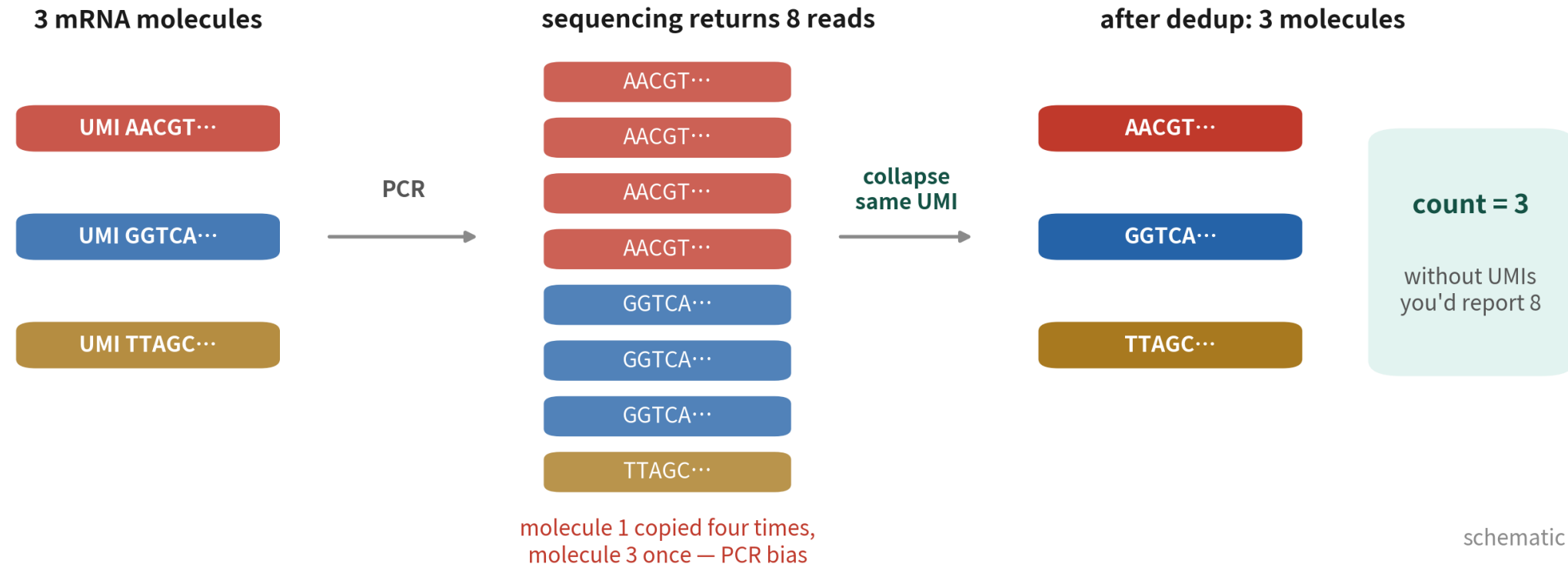
schematic

Smart-seq tiles the whole transcript; 10x 3' reads only the end — this cut decides what analyses you can ever do

UMI deduplication: folding reads back into molecules

UMI deduplication: folding reads back into molecules

PCR amplifies unevenly, so read counts lie; UMIs let you count molecules



schematic

with UMIs, counts are molecules; without (Smart-seq2), counts are reads — different physical scales

With UMIs counts are molecules; without (Smart-seq2) counts are reads — different physical scales

The other two families

Microwell: Seq-Well



cells settle into a well array by gravity — no precision pumps, portable and low-tech

Combinatorial: sci-RNA-seq3



barcode first, pool later; rounds of barcoding multiply — scale reaches millions

Plate barcode: MARS-seq / CEL-seq2



plate sorting + per-well barcodes + pooled libraries; UMIs, 3'-end reads

The four families, one table

Scales and depths are orders of magnitude — exact numbers live on official pages

Family	Isolation	UMIs	Reads	Scale per run
Droplet (10x, Drop-seq, inDrops)	oil droplets + barcoded beads	yes	3' end	thousands to tens of thousands
Microwell (Seq-Well)	gravity into microwells	yes	3' end	thousands to tens of thousands
Plate full-length (Smart-seq2/3)	sorted into plates, one cell per well	SS2: no	full-length	dozens to thousands
Combinatorial (sci-RNA-seq3 etc.)	barcode first, pool, multiply	yes	3' end	hundreds of thousands to millions

THE CORE MNEMONIC

**To place a platform, ask three things:
how it isolates, whether it has UMIs, what it reads.**

Those three answers determine what the data looks like, what QC should expect, and what downstream can do.

02

Platform profiles

Each platform pairs with an exercise dataset — a profile is not a spec sheet, it predicts what your data will look like

10x Chromium: the default answer

The thickest ecosystem



Cell Ranger through
Seurat, end to end —
B1 to B5 ran on it

Scale: thousands to tens of thousands



the 5k PBMC of
exercise Basic-01
(~5,000 cells) is one
standard run

The price



3' end only, shallow
per cell; doublets and
ambient RNA are
droplet original sins
(Basic-10)

Smart-seq2: the small, deep microscope

Full-length, deep



per-cell reads on the order of hundreds of thousands; low-expression genes, isoforms, coding mutations all visible

Scale: hundreds to thousands



Darmanis GBM 3,589 cells, Tirosh melanoma 4,645 — the tumour classics ran on it

No UMIs



counts are reads and length-biased; normalization and QC need a different head (section 3)

Drop-seq vs inDrop: the academic forerunners

Two 2015 droplet systems — the academic prototypes 10x commercialized

Drop-seq

disposable, cheap, build-your-own

- exemplar: Basic-08 mouse retina, 44,808 cells
- bead capture is stochastic; cell utilization low
- the workhorse of the first atlas era

inDrop

hydrogel beads, controlled encapsulation

- exemplar: Basic-02 Baron islets, ~8,569 cells
- higher capture — kinder to scarce samples
- shallow per cell; lower your QC expectations

MARS-seq vs CEL-seq2: plates with barcodes

Plate-sorting precision plus pooled-barcode efficiency — the family between plates and droplets

MARS-seq

384-well plates + sorting

- exemplar: Basic-04 Paul haematopoiesis, ~2,730 cells
- Advanced-19 myeloma, 51,840 cells — stack plates and scale grows
- the signature pipeline of immunology labs

CEL-seq2 / SORT-seq

IVT linear amplification + pooled barcodes

- exemplar: Basic-09 Muraro islets (4 donors)
- SORT-seq is CEL-seq2, automated
- matrices ship with failed wells — check `dim()` on load

Seq-Well: single-cell to go

Pump-free microwells



light equipment, low cost — feasible at the bedside or in the field

Droplet-class scale



Advanced-05 AML
~38,410 cells;
Advanced-16 prostate (S3) 21,743 cells

Data that looks like 10x



UMIs + 3' end; the Basic-06 COVID cohort ships as 10x-style matrices

sci-RNA-seq3: the million-cell world

Barcodes that multiply



rounds of
combinatorial
indexing, no per-cell
chamber — scale
grows exponentially

Exemplars: two developmental atlases



Advanced-56 MOCA,
1,331,984 cells after
official cleaning;
Advanced-57 human
fetal atlas, ~4 million

The price: very shallow cells

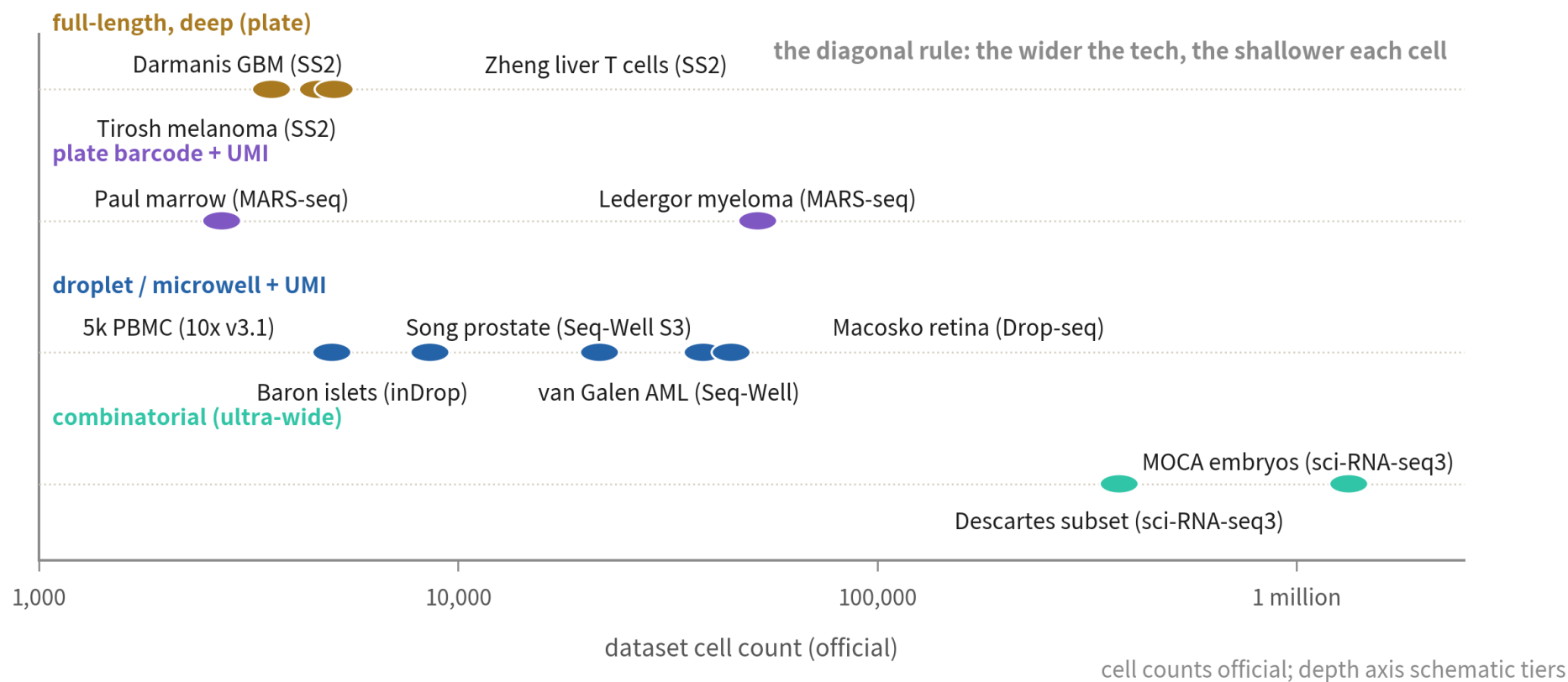


statistical power
comes from cell
numbers; per-cell
claims must be
conservative

The exercise bank, mapped: cells × depth

Real datasets from the exercises: cell count vs depth tier

x = official cell counts (log scale); y = per-cell depth tier (schematic), deeper upward



Cell counts are official, depth is by tier — the diagonal rule: the wider the tech, the shallower each cell

Platform profiles at a glance

Every exemplar comes from the exercise bank — pick a platform, follow the exercise number

Platform	Family	UMIs	Exemplar (exercise)	Official cells
10x Chromium 3'	droplet	yes	5k PBMC (Basic-01)	~5,000
Smart-seq2	plate full-length	no	GBM core/margin (Adv-02)	3,589
Drop-seq	droplet	yes	mouse retina (Basic-08)	44,808
inDrop	droplet	yes	Baron islets (Basic-02)	~8,569
MARS-seq	plate barcode	yes	haematopoiesis (Basic-04)	~2,730
CEL-seq2 / SORT-seq	plate barcode	yes	Muraro islets (Basic-09)	4 donors
Seq-Well	microwell	yes	van Galen AML (Adv-05)	~38,410
sci-RNA-seq3	combinatorial	yes	MOCA embryos (Adv-56)	1,331,984

HOW TO USE A PROFILE

**Meet the data, place the platform —
the profile tells you what to expect.**

What magnitude nCount should be, which cell types tend to go missing, which analyses are feasible — settle that first and QC stops lying to you.

03

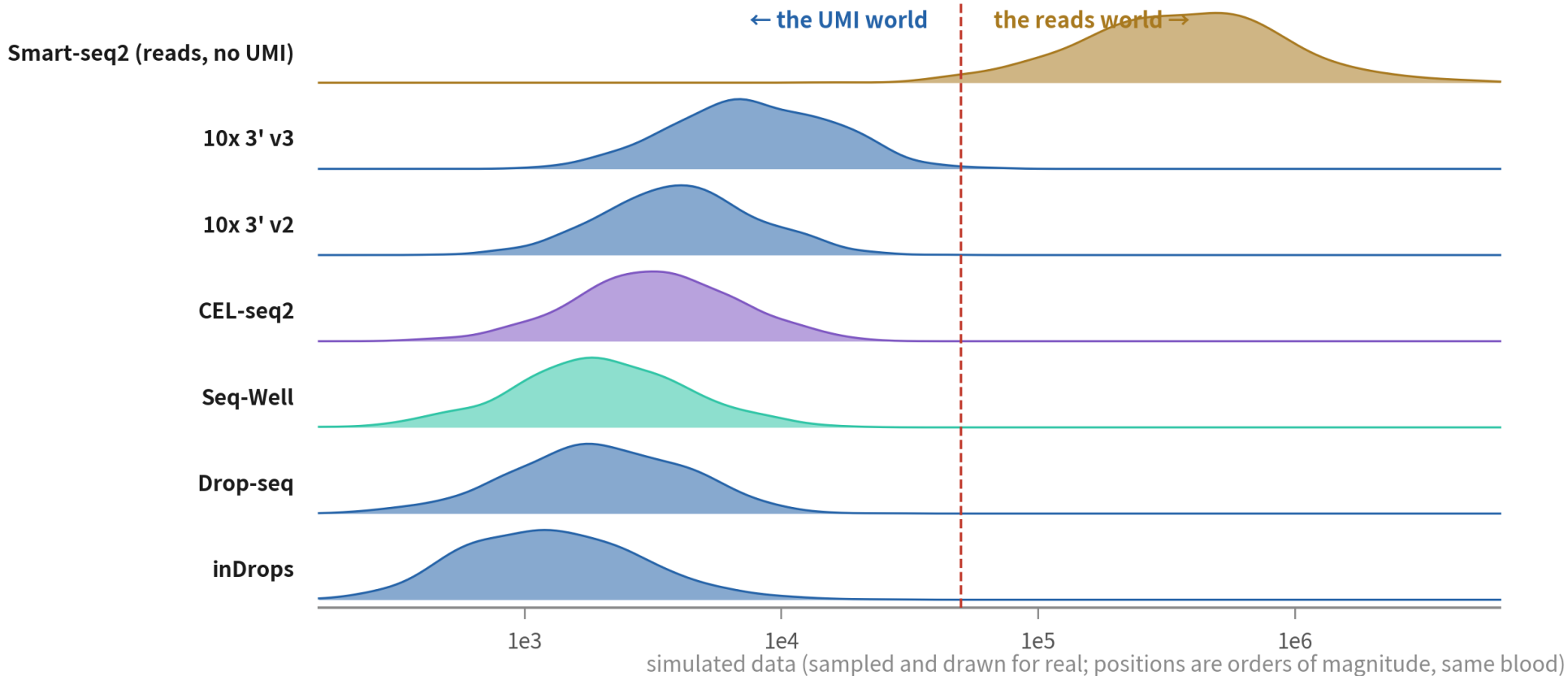
The platform dictates QC and analysis

Same functions, different expectations — the reasons behind thresholds live in the platform

nCount across technologies

nCount magnitude: seven technologies, not one number line

Droplet and microwell count UMIs; Smart-seq2 counts reads — ask the unit before the size



The cold open's answer: Smart-seq2 counts reads — one to two orders of magnitude above UMIs by nature

Normalizing without UMIs: length enters

The hidden variable without UMIs: gene length

Two genes, three molecules each — equal with UMIs; several-fold apart in reads

long gene (say 10 kb)



molecules: 3

short gene (say 1 kb)



full-length reads: 12

molecules: 3

full-length reads: 3

same 3 molecules

UMI counts
3 vs 3 ✓

read counts
12 vs 3 ✗

the fix: CP10K + log1p as usual for UMI data; length-correct (TPM-style) non-UMI data before comparing across genes

schematic (read tallies illustrative)

- UMI data: molecules counted regardless of length — CP10K + log1p and you're set (B6)
- read data: long genes pile up more reads — length-correct (TPM-style) before comparing across genes
- which is why Smart-seq classics often ship as TPM matrices (Advanced-04 is TPM)

Full-length vs end-counting: the downstream divide

What gets read isn't a detail — it's which questions you can ever ask

Full-length (Smart-seq)

sequence-rich

- isoforms and alternative splicing visible
- coding mutations detectable — B16 noted 10x's 3'-only handicap
- full-length TCR/BCR reconstruction possible

End-counting (10x 3')

clean counts at scale

- UMI dedup absorbs PCR bias
- paired TCR/BCR needs 10x 5' with V(D)J (Adv-38, 45)
- most sequence-level questions stay out of reach

One tool, two platforms: inferCNV's cutoff

```
cnv <- infercnv::run(cnv,  
  cutoff = 1,      # Smart-seq: 每細胞計數深, 用 1  
  out_dir = "output/bs2_infercnv",  
  denoise = TRUE)
```

```
# 換成 10x 資料, 同一行要改:  
# cutoff = 0.1      # 差十倍——理由不在工具,  
#                  # 在平台的每細胞深度 (B16)
```

B16's exact warning: the wrong platform's line either guts your genes or keeps a pile of noise

The platform checklist to run first

Five questions — each one changes your next move

Check	Why	Action
Are counts UMIs or reads?	one to two orders apart; QC misreads without it	identify the platform; confirm with the distribution
What magnitude should nCount be?	each platform has its own 'normal'	look first, threshold second (B5)
Does normalization need length?	read counts without UMIs carry length bias	UMI: CP10K; no UMI: TPM-style
Does the tool have a platform dial?	e.g. inferCNV cutoff 1 vs 0.1	read the docs; don't copy numbers
What is this platform's original sin?	droplets: doublets/ambient; plates: well effects and failed wells	targeted QC (Basic-10, Basic-09)

SECTION 3, THE IRON RULE

QC thresholds never move between platforms

—

rules travel; numbers do not.

B5's 'look at the distribution, set the cutoff, write down why' works on every platform,
unchanged.

04

The experiment: one blood, seven technologies

pbmcsca — biology pinned down, what remains is technology

Loading pbmcscsca

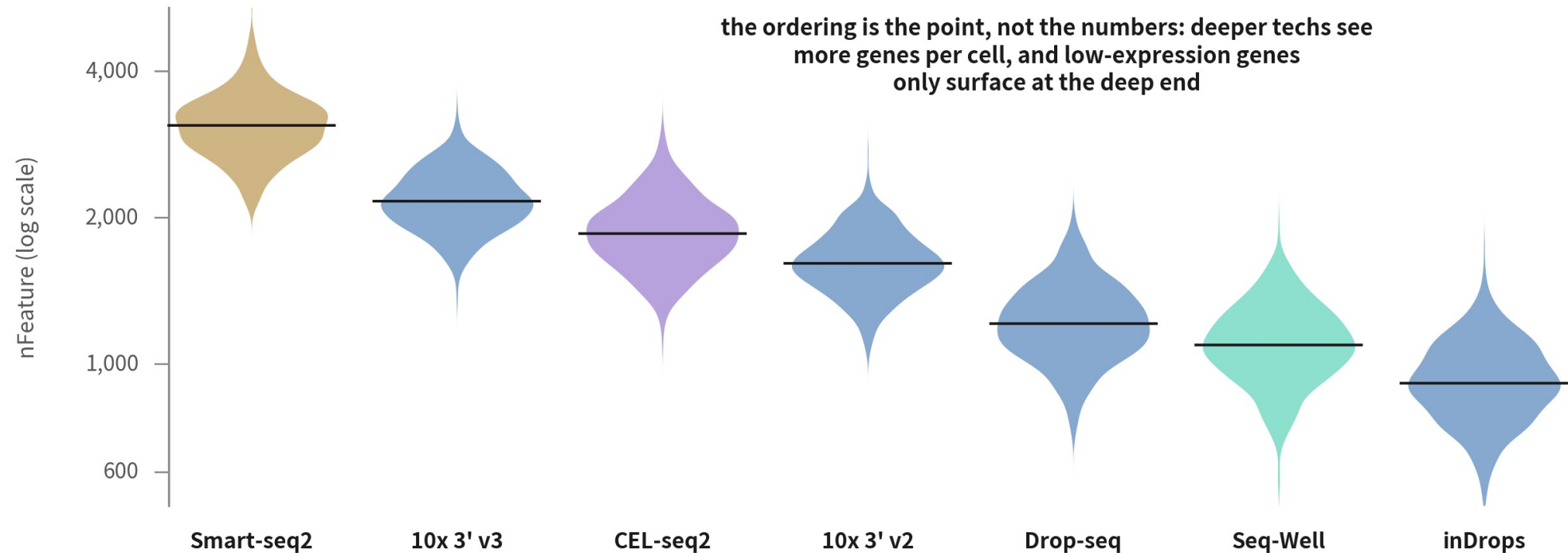
```
install.packages("SeuratData",  
                 repos = "https://seurat.nygenome.org")  
library(SeuratData)  
InstallData("pbmcscsca") # Ding 2020 ; GSE132044  
pbmcscsca <- LoadData("pbmcscsca")  
pbmcscsca # 31,021 cells  
  
colnames(pbmcscsca[[[]]) # 欄名親眼確認，別抄教學
```

One PBMC sample run on seven technologies: 10x v2/v3, Smart-seq2, Seq-Well, inDrops, Drop-seq, CEL-seq2

Seven technologies: genes per cell

One blood sample, seven technologies: genes detected per cell

Biology is pinned; what remains is technology — the pbmcsc design



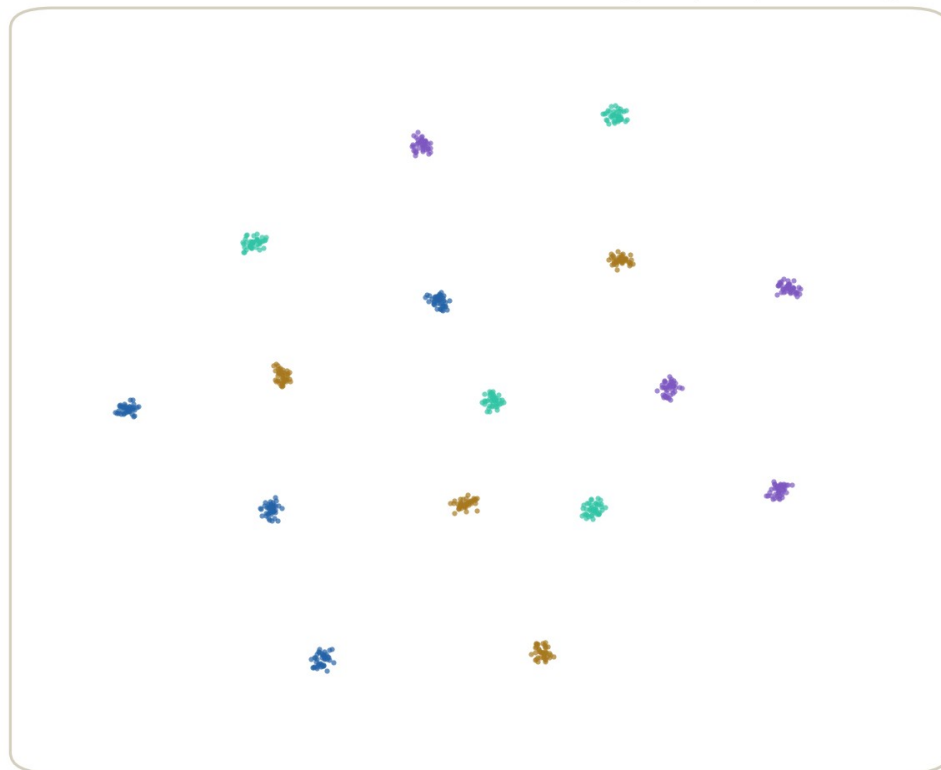
simulated data (sampled for real; schematic magnitudes, not measured pbmcsc values)

The ordering is the point: deeper techs see more genes — and normalization cannot resurrect the undetected

Unintegrated vs integrated

Look at the technology effect raw — then see what integration recovers

Same simulated cells embedded twice: left groups by tech, right aligns by type



unintegrated, coloured by technology: continents



simulated data (PCA + knn embedding, both actually run)

integrated, coloured by cell type: aligned

Unintegrated, cells group by technology (B11's tech effect); integrated, types align across technologies

A cell type missing on one platform — three hypotheses

Can't detect it



sensitivity or cell-size
preference — the tech
is blind to it by design

The workflow lost it



sorting, lysis, settling
— the cells died before
the machine

Not enough sampling

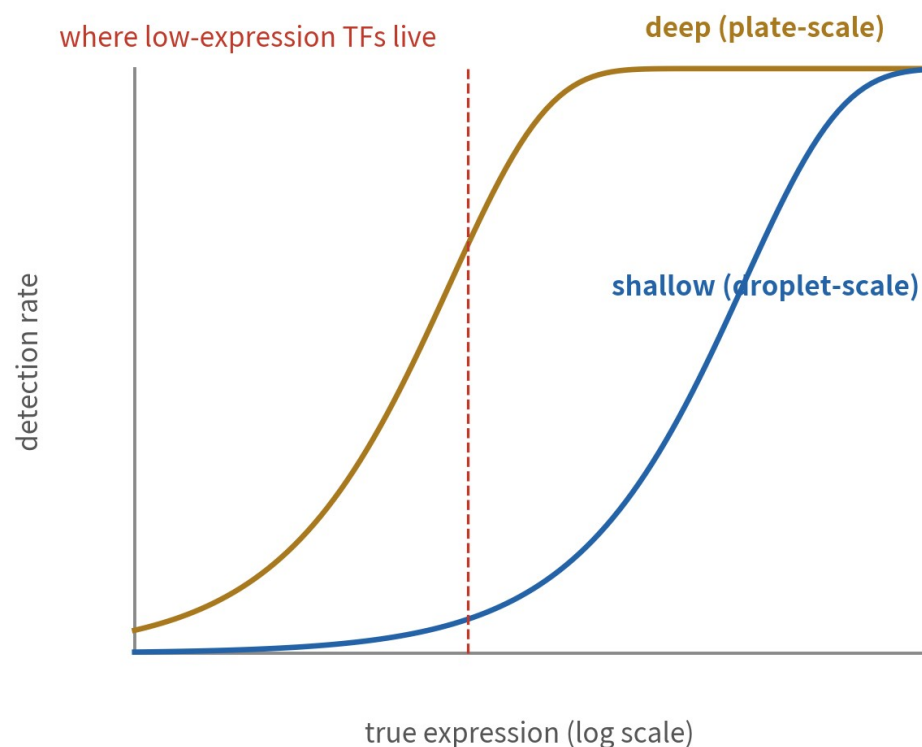


too few cells; the rare
population simply
wasn't drawn — more
cells brings it back

Depth versus breadth

Depth versus breadth: two microscopes, two kinds of truth

Left: only depth reveals low-expression genes (Poisson, computed). Right: only breadth counts rare populations (computed)



a 1% population: how many cells do you net?

deep & small: 5,000 cells



expect 50 cells

wide & shallow: 50,000 cells



expect 500 cells

**50 cells won't form a cluster; 500 will —
rare populations are bought with cell numbers**

simulated data (both panels computed)

Advanced-09's lesson: 5,063 Smart-seq2 T cells saw low-expression regulators that tens of thousands of 10x cells could not

Deep-and-small vs wide-and-shallow

Two microscopes see two truths — the research question picks the instrument

Deep & small

the plate route

- low-expression TFs and receptors visible
- isoforms, mutations, full-length receptor sequences
- rare populations easily missed wholesale

Wide & shallow

the droplet / combinatorial route

- rare populations netted by cell numbers
- the right answer for composition and atlas-scale questions
- per-cell signal is fuzzy; interpret conservatively

THE POINT OF THIS EPISODE

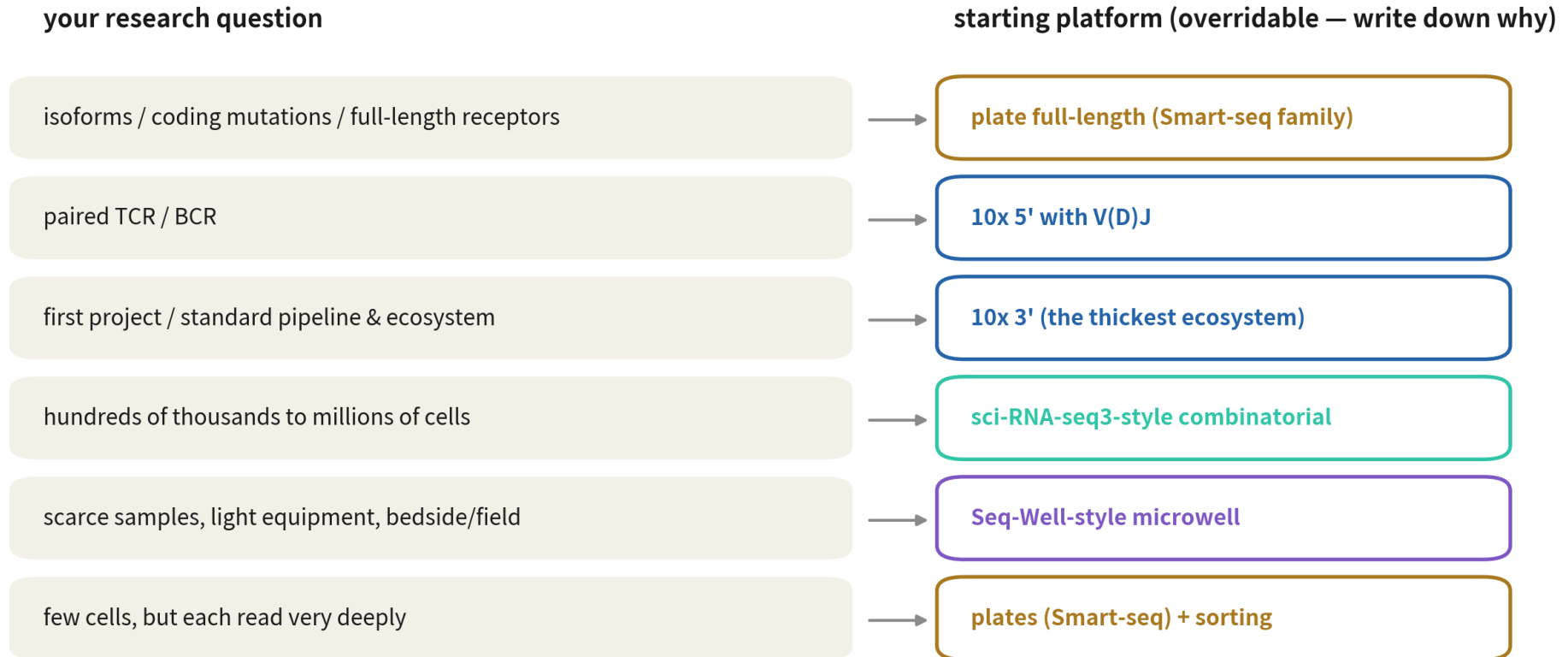
**The platform is part of your hypothesis:
it decides what you can and cannot see.**

One level above 'every parameter is a hypothesis' — even the machine that made the data is part of the analysis.

The platform decision table

The platform decision table: start from the question

Walk forward to choose a platform; walk backward on public data — identify the platform, then set expectations



schematic

Walk it forward to choose; walk it backward on public data — identify the platform, then set expectations

This episode in four lines

If you can justify the right column, the episode is yours

Section	One line
1. The family tree	three questions place any platform: isolation, UMIs, what's read
2. Profiles	profiles predict the data; every platform has a real dataset in the bank
3. QC logic	thresholds don't travel; walk unit, magnitude, length, parameters
4. The experiment	tech effects are bigger than you think; depth-for-breadth is a real trade

Check yourself 1

Same blood: 10x reports median nCount in the thousands, Smart-seq2 in the hundreds of thousands. First move?

- A Suspect the Smart-seq2 machine
- B Check the counting unit on each side: UMIs vs reads
- C Downsample Smart-seq2 to match 10x
- D Merge both and cluster

Answer B. Smart-seq2 has no UMIs — its counts are reads, one to two orders above UMI counts by nature. That's physics, not a fault. Until the unit is confirmed, every QC call is a guess.

Check yourself 2

Why does non-UMI data need length correction (TPM-style) before comparing across genes?

- A Because its sequencing quality is worse
- B Because long genes pile up more reads — counts mix in a length effect
- C Because TPM is the newer method
- D It doesn't: log-normalization fixes length

Answer B. With full-length reads, the same three molecules yield more fragments from a long gene — length leaks into the counts. UMIs count molecules and are immune; without them you must divide length out. Log-normalization fixes depth, not length.

Check yourself 3

You study low-expression TFs in T cells, plus their mutation status, with budget for a few thousand cells. Which way?

- A 10x 3' — the fullest ecosystem
- B sci-RNA-seq3 — the most cells
- C Plate full-length (Smart-seq) + sorting
- D Seq-Well — the cheapest equipment

Answer C. Both needs point to full-length depth: low-expression genes demand depth, and mutations require reading the coding body. Few cells is fine — this is depth-for-breadth used well, exactly how Advanced-09's 5,063 T cells were done.

NEXT EPISODE

BS3: The tool landscape

Platform chosen — now which tools? Over eighteen hundred of them, still growing.

Next episode: a map and a four-step method. Want hands-on first? Exercise Basic-11 is this episode in practice.